



Humber ASV Document Title

Humber Polytechnic

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A Glossary

Autonomous Surface Vehicle (ASV) An autonomous surface vehicle (ASV) is an unmanned vessel that operates on the sea surface without real-time input or control from human operators.[5]

Base Station A fixed radio communications station that can communicate with mobile radio devices in its neighbourhood. Forms of radio communication that have base stations include radio wireless data and mobile telephony...[1]

Loon-E Humber ASV's current ASV.[**HuMBERASVTEAM**]

HumberASV Humber Polytechnic's Mechatronics club. A multidisciplinary team of passionate Humber Polytechnic students and professionals dedicated to advancing autonomous maritime technology.[2]

LIDAR LiDAR, an acronym for "light detection and ranging," is a remote-sensing technology that uses laser beams to measure precise distances and movement in an environment, in real time.[3]

Robot Operating System (ROS) The Robot Operating System (ROS) is a set of software libraries and tools that help you build robot applications.[4]

B Bibliography

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- [3] IBM. *What is LIDAR?* 2026. URL: <https://www.ibm.com/think/topics/lidar> (visited on 03/17/2026).
- [4] Open Robotics. *ROS*. 2026. URL: <https://www.ros.org/> (visited on 03/17/2026).
- [5] Arthur Trembanis, Mark Lundine, and Kaitlyn McPherran. "Coastal Mapping and Monitoring". In: *Encyclopedia of Geology (Second Edition)*. Ed. by David Alderton and Scott A. Elias. Second Edition. Oxford: Academic Press, 2021, pp. 251–266. ISBN: 978-0-08-102909-1. DOI: <https://doi.org/10.1016/B978-0-12-409548-9.12466-2>. URL: <https://www.sciencedirect.com/science/article/pii/B9780124095489124662>.